
$\widetilde{U}_i^{\mathrm{FP}}$ in chronological order
$|\widetilde{\Psi}_i\rangle$
$\widetilde{\mathcal{S}}^{(\eta_i)}_{i+1,\beta_{i,1}}$
$\widetilde{\mathcal{S}}^{(\eta_i)}_{i,-\alpha_{i,1}}$
$\cdots$
$\widetilde{\mathcal{S}}^{(\eta_i)}_{i+1,\beta_{i,r_i}}$
$\widetilde{\mathcal{S}}^{(\eta_i)}_{i,-\alpha_{i,r_i}}$
$|\widetilde{\Psi}_{i+1}\rangle$